Implementation of the Characteristic Value Iteration Algorithm

Davide Baldelli
Mila & Polytechnique Montréal
davide.baldelli@mila.quebec

Thomas Mousseau
Mila & Polytechnique Montréal
thomas.mousseau@mila.quebec

Abstract

Distributional Reinforcement Learning (DistRL) aims to model the full return distribution, rather than just the expected value. While current state-of-the-art methods like C51 and QR-DQN rely on heuristic projections to approximate the distributional Bellman update, Characteristic Value Iteration (CVI) offers a theoretically elegant alternative by operating in the frequency domain, where the convolution of random variables becomes a simple element-wise multiplication. In this work, we bridge the gap between the theoretical formulation of CVI and its practical application. By providing a rigorous validation in the tabular setting and extending this analysis to the approximate setting via CF-DQN, we establish CVI as a viable, projection-free framework for DistRL.

Tabular CVI: **github.com/Dundalia/CVI**
Approximate CVI: **github.com/Dundalia/CF-DQN**

1 Introduction

Reinforcement Learning has traditionally focused on maximizing the expected cumulative return, effectively condensing the randomness of the environment into a single scalar mean. While this approach has driven significant successes, the expected value often fails to capture critical information about the underlying uncertainty of returns. In risk-sensitive domains, such as autonomous driving and robotics, where tail events can lead to catastrophic behaviors, knowing the full probability distribution of returns is often more valuable than knowing the mean alone.

Distributional Reinforcement Learning (DistRL) addresses this limitation by modeling the complete distribution of the random return, rather than just its expectation. This richer representation allows agents to distinguish between actions with similar expected returns but vastly different variances, enabling safer and more robust policy learning.

Most existing DistRL algorithms, such as C51 (Bellemare et al., 2017) or QR-DQN (Dabney et al., 2018), approximate this distribution in the probability domain. However, these methods often require categorical or heuristic projection steps and fixed domain to handle the Bellman update. In contrast, Farahmand (2019) presented a theoretical framework proposing a fundamentally different approach, representing the value distribution in the frequency domain using its Characteristic Function (CF).

This method, named **Characteristic Value Iteration (CVI)**, leverages the property that the sum of independent random variables corresponds to the element-wise product of their characteristic functions. This transforms the complex convolution required by the distributional Bellman operator into a simple multiplication. Despite the theoretical elegance of this approach, the original work focused on establishing the mathematical foundations, including the contraction properties of the operator in the frequency domain, and did not provide an empirical evaluation.

The goal of this project is to bridge this gap between theory and practice. We provide the first implementation of the CVI algorithm to verify its theoretical properties and evaluate its practical efficacy compared to established DistRL baselines.

In summary, our contributions are as follows:

- Implementation of CVI in a tabular setting to empirically validate the theoretical assumptions and Bellman contraction properties.
- Implementation of Approximate CVI in a high-dimensional setting using function approximation on four games from the Atari benchmark, and comparison against established baselines (C51 and DQN).

2 Prior Work

Categorical Deep Q-Network (C51), introduced by Bellemare et al. (2017), provided the first state-of-the-art results demonstrating the benefits of learning a value distribution instead of a function. In C51, the distribution of returns is modeled as a categorical distribution over a fixed, discrete support. The algorithm defines a set of N atoms (typically $N = 51$) uniformly spaced between a minimum and maximum value (V_{min} and V_{max}). The network outputs the probabilities associated with each atom. However, C51 suffers from the requirement of knowing the bounds of the return ($V_{\text{min}}, V_{\text{max}}$) making it bound to express all of its probability mass inside of these constraints regardless training results.

To address the limitations of the fixed support in C51, Dabney et al. (2018) propose Quantile Regression DQN (QR-DQN). Their work is built on top of a standard Deep Q-Network (Mnih et al., 2015), which uses a deep neural network to approximate the action-value (Q) function. Instead of fixing the span of the distribution and learning the probabilities, QR-DQN fixes the probability masses to be uniform and learns the quantile locations of the cumulative distribution function. This approach allows the agent to model distributions with arbitrary support bounds and generally results in more accurate approximations of the return distribution compared to C51, without requiring manual tuning of distribution limits.

This work was later extended by IQN (Dabney et al., 2018), which generalizes by learning an implicit quantile function, conditioned on sampled quantile fractions. Similarly, the QR-DQN approach was combined with other DQN extensions to create Rainbow (Hessel et al., 2018). Rainbow is a unified high-performing DQN agent, with 8 previously proposed extensions combined, including DistRL. Finally, FQF (Yang et al., 2019) is a more recently proposed extension that further improves QR-DQN by learning the quantile fractions themselves through a dedicated fraction proposal network.

3 Background

3.1 Distributional RL

Distributional Reinforcement Learning considers a discounted Markov Decision Process (MDP) $(\mathcal{X}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$. In standard RL, the goal is to maximize the expected return, however DistRL is interested in the random variable or the return itself.

$$G^\pi(x) \overset{D}{=} R_0 + \gamma G^\pi(X') \tag{1}$$

where $X' \sim \mathcal{P}(\cdot|x, \pi(x))$ and the equality holds in distribution.

Finding the distribution of G^π involves handling the sum of the reward R and the discounted next-state return $\gamma G^\pi(X')$. In the probability domain (used by methods like C51 and QR-DQN), the probability density function (PDF) of a sum of independent random variables is the **convolution** of their individual PDFs, $p_G(x) = (p_R * p_{\gamma G'})(x)$. Computing this convolution is computationally expensive and difficult to approximate, thus prior work (Bellemare et al., 2017; Dabney et al., 2018) often relied on the heuristic projection steps or quantile approximations.

3.2 Characteristic Functions in RL

To address the limitations of convolution and projection, Farahmand (2019) proposes modeling the return distribution in the frequency domain. The Characteristic Function (CF) of a random variable

X is defined as

$$c_X(\omega) \triangleq \mathbb{E}[e^{i\omega X}] = \int_{x \in X} \exp(jx\omega)\mu(dx) \qquad \omega \in \mathbb{R} \tag{2}$$

We define the **Characteristic Value Function (CVF)**, $\tilde{V}^\pi(\omega; x)$, as the CF of the return $G^\pi(x)$. A key property of characteristic functions is that the CF of a sum of independent random variables is the element-wise product of their individual CFs. This transforms the distributional Bellman equation into:

$$\tilde{V}^\pi(\omega; x) = \mathbb{E}\left[e^{i\omega R}\tilde{V}^\pi(\gamma\omega; X')\right] \tag{3}$$

3.3 The Bellman Operator in Frequency Domain.

As proven by Farahmand (2019), the Bellman operator in the frequency domain ($\tilde{T}$) is a contraction with respect to a weighted L_∞ norm. This contraction property suggests that one can find the true CVF through an iterative procedure similar to standard Value Iteration. Consequently, Farahmand (2019) proposes an empirical risk minimization approach.

$$\hat{L}_n(\theta) = \frac{1}{n}\sum_{k=1}^{n}\int w(\omega)\left|\tilde{V}_\theta(\omega; X_k) - \left(e^{i\omega R_k}\tilde{V}_{\theta'}(\gamma\omega; X'_k)\right)\right|^2 d\omega \tag{4}$$

where $w(\omega)$ is a weighting function that focuses the error minimization on important frequencies and $\tilde{V}_{\theta'}$ represents the target network.

4 Control in Frequency Space

4.1 Tabular Implementation and Design Space

The practical implementation of CVI in tabular MDPs requires three fundamental design decisions that emerge from the discretization of the continuous frequency domain.

First, we must select a **frequency grid** $\Omega = \{\omega_1, \ldots, \omega_K\} \subset [-W, W]$ on which to evaluate and store the characteristic function, as maintaining $\tilde{V}(\omega; x)$ for all $\omega \in \mathbb{R}$ is computationally intractable. Second, the CF Bellman operator (Equation 4) requires evaluating $\tilde{V}(\gamma\omega; x')$at scaled frequencies that typically do not coincide with grid points, necessitating an **interpolation method** to estimate off-grid CF values from the discrete representation. Third, to enable greedy policy improvement, we must **collapse** the distributional representation $\tilde{V}(\omega; x, a)$ back to scalar Q-values $Q(s, a) = \mathbb{E}[G(s, a)]$, which corresponds to extracting the first moment from the characteristic function via the relationship $\mathbb{E}[G] = -i\tilde{V}'(\omega; x)$.

To systematically explore this design space, we evaluate seven distinct grid construction strategies that vary in their density allocation across the frequency range (see Figure 1): **(i)** uniform grids with constant spacing. **(ii-iv)** Multi-density region grids that concentrate points near $\omega = 0$ with two, three, or four progressive density tiers. **(v-vii)** Decay-based grids following exponential, linear, or quadratic density functions that smoothly decrease point density toward the tails. The multi-density strategies are motivated by the observation that moment extraction occurs at $\omega = 0$, suggesting concentrated sampling in this region, while decay-based approaches provide smooth transitions without sharp density boundaries. This search was performed using the `Taxi-v3` environment.

For interpolation, we implement four methods that differ in their treatment of the complex-valued CF. **(i)** Linear interpolation applies standard piecewise linear functions independently to real and imaginary components. **(ii)** Polar interpolation separates magnitude $|\tilde{V}(\omega)|$ and unwrapped phase $\arg(\tilde{V}(\omega))$, interpolating each independently before reconstruction, which better preserves the validity constraint $|\tilde{V}(\omega)| \leq 1$. **(iii)** PCHIP (Piecewise Cubic Hermite Interpolating Polynomial) applies monotonicity-preserving cubic interpolation to real and imaginary parts (Fritsch and Carlson, 1980), avoiding Runge oscillations while maintaining smoothness. **(iv)** Lanczos interpolation employs windowed

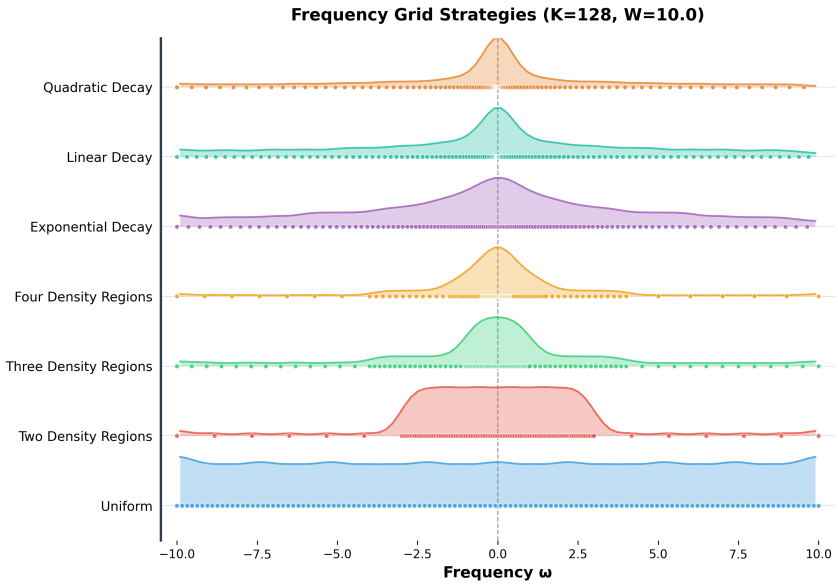


Figure 1: **Frequency grid construction strategies for Characteristic Value Iteration.** Visualization of seven grid strategies distributing $K = 128$ points over the frequency range $[-W, W]$ with $W = 10$. Each row shows the discrete point locations (vertical lines) and the corresponding point density function (shaded region).

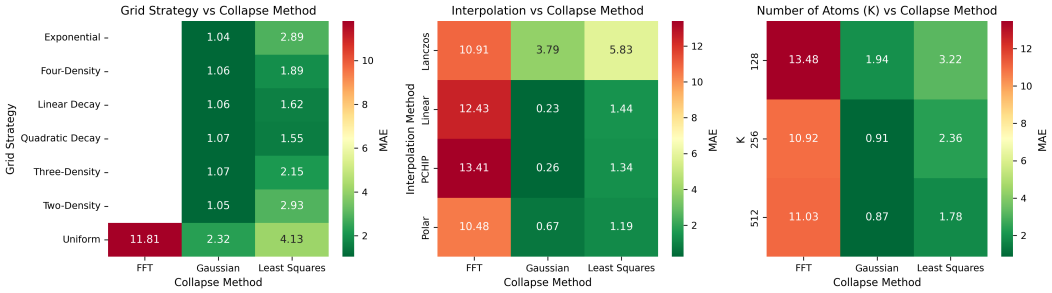


Figure 2: Heatmap plots displaying the results of our *grid search* over grid, interpolation, and collapse design parameters. The combined best option was **Three-Density, Polar,** and **Gaussian**.

kernels (Duchon, 1979), which are theoretically optimal for bandlimited signals on uniform grids but computationally more expensive.

Finally, we consider three collapse methods for mean extraction. **(i)** Least-squares fitting approximates $\tilde{V}(\omega)$ locally around $\omega = 0$ with a quadratic polynomial $a_0 + a_1\omega + a_2\omega^2$, extracting the mean as $\mathbb{E}[G] = \mathrm{Im}(a_1)$ from the linear coefficient. **(ii)** FFT-based inversion applies the inverse Fourier transform to recover the probability density in the spatial domain, computing the mean directly as $\int xp(x)dx$. **(iii)** Gaussian approximation assumes a locally Gaussian CF structure $\log \tilde{V}(\omega) \approx i\mu\omega - \frac{1}{2}\sigma^2\omega^2$, unwrapping the phase and fitting a line through the origin to extract μ as the slope. Each method represents a different trade-off between computational complexity, assumptions about the CF structure, and robustness to discretization artifacts.

These parameters were explored using a grid search to find the final optimal parameters. The results of the grid search are shown in Figure 2.

4.2 Function Approximation with Neural Networks

Motivated by the empirical findings from our tabular experiments, we extend CVI to high-dimensional state spaces via deep neural networks. Our architecture, which we term **CF-DQN**, directly param-

eterizes the characteristic function $\tilde{V}_\theta(\omega; x) : \mathcal{X} \times \mathbb{R} \to \mathbb{C}$ as the output of a convolutional neural network. Following the validated design principles from Section 5.1, we adopt a three-density-region frequency grid strategy that allocates 50% of the K grid points to the inner 10% of the frequency range $[-W, W]$, 30% to the middle region, and the remaining 20% to the tails. For the critical interpolation step required to evaluate $\tilde{V}(\gamma\omega; x')$ at scaled frequencies, we exclusively use polar interpolation, which separates magnitude and unwrapped phase components to preserve the validity constraint $|\tilde{V}(\omega)| \leq 1$. To extract scalar Q-values for action selection, we employ the Gaussian collapse method, which fits a linear model to the unwrapped phase in the low-frequency region $|\omega| \leq 2.0$, exploiting the local Gaussian structure of bounded-reward return distributions.

We implement CF-DQN within the CleanRL framework (Huang et al., 2022), adopting the standard DQN architecture (Mnih et al., 2015) with three convolutional layers followed by fully-connected layers. The network outputs $2K$ real values per action, which we interpret as the real and imaginary components of the characteristic function at the K grid points. Critically, we introduce a soft regularization term $\lambda_{\phi_0}\mathbb{E}\left[|\tilde{V}(0; x, a) - 1|^2\right]$ to encourage the network to respect the fundamental CF property $\tilde{V}(0) = 1$, which is essential for accurate moment extraction. Our training objective combines the standard complex-valued L_2 loss in the frequency domain with this validity penalty:

$$\mathcal{L}(\theta) = \mathbb{E}_{(s,a,r,s')\sim\mathcal{D}}\left[\|\tilde{V}_\theta(x, a) - \tilde{T}\tilde{V}_{\theta^-}(x')\|_2^2\right] + \lambda_{\phi_0}\mathbb{E}\left[|\tilde{V}_\theta(0; x, a) - 1|^2\right], \tag{5}$$

where $\tilde{T}$ denotes the CF Bellman operator (Equation 4), θ^- are the target network parameters, and $\mathcal{D}$ is the replay buffer. This formulation sidesteps the complex categorical projection step required by C51 (Bellemare et al., 2017), replacing it with a straightforward regression objective that naturally operates in the frequency domain. We employ standard DQN techniques including target networks (updated every $10{,}000$ steps), experience replay (buffer size 10^6), and ϵ-greedy exploration with linear annealing. All other hyperparameters follow the established baselines for Atari 2600 benchmarks, ensuring a fair comparison with prior distributional RL methods.

5 Results

5.1 Tabular Setting

Before applying Characteristic Value Iteration (CVI) to high-dimensional problems with function approximation, we first validate the algorithm in a tabular setting. Our primary goals are to ensure that CVI converges to the same optimal expected value as classical planning algorithms, such as Value Iteration and Policy Iteration, and to verify that CVI can accurately recover the full distribution of returns in a stochastic environment, comparing it against ground-truth Monte Carlo estimations and the C51 baseline. All of the following experiments were ran using the same default configuration to ensure a fair comparison 1.

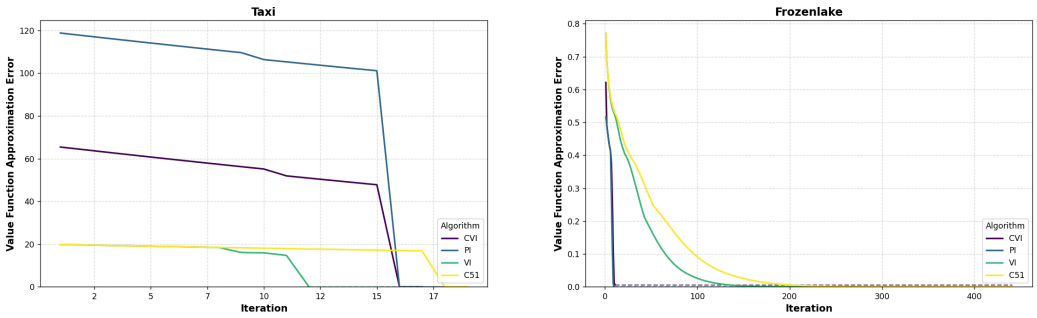
5.1.1 Convergence Analysis: Taxi-v3

We utilized the `Taxi-v3` from Gymnasium with deterministic dynamics to test the algorithm on a medium-sized state space ($|\mathcal{S}| = 500$). The primary metric used to measure performance is the mean value of all states over time. We initialized each algorithm at different starting states, as they are expected to converge to the same final mean value function regardless.

As shown in Figure 3(a), we compared CVI against Value Iteration (VI), Policy Iteration (PI), and C51. The results demonstrate that CVI (blue) successfully converges to the same asymptotic value as the classical dynamic programming methods (VI and PI), regardless of their random initialization. This empirically validates the contraction property of the Bellman operator in the frequency domain. Despite operating on complex-valued characteristic functions, the derived mean value remains consistent with the expected theory.

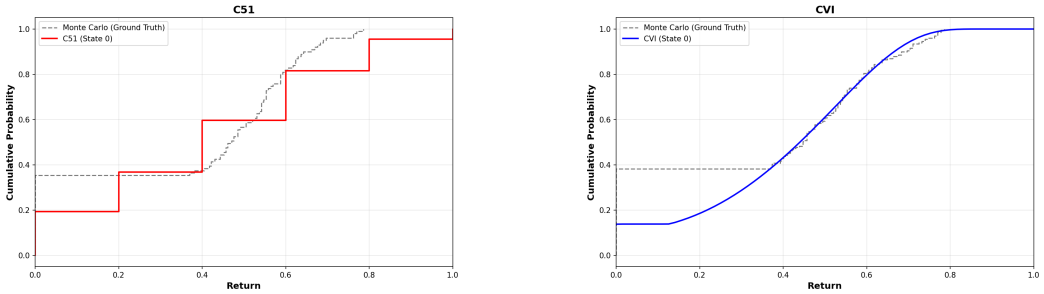
5.1.2 Distributional Analysis: Stochastic FrozenLake

To evaluate the algorithm's capability to model complex distributions, we utilized the `FrozenLake-v1 8x8` environment from Gymnasium with stochastic dynamics enabled



(a) (b)

Figure 3: **Value Function Approximation Error on Taxi-v3 (a) and FrozenLake (b).** Comparison of CVI, VI, PI, and C51 relative to a high-precision ground truth ($10,000$ iterations, $\epsilon = 10^{-12}$). In both deterministic and stochastic settings, CVI successfully reduces the approximation error to almost zero, converging to the same optimal fixed point as standard planning algorithms.



(a) (b)

Figure 4: **CDF Comparison in State 0.** (a) C51 produces a step-function CDF due to its reliance on a fixed, discrete support. (b) CVI produces a smooth, continuous approximation that closely follows the Monte Carlo ground truth, demonstrating the representational flexibility of the frequency domain. The observed discrepancy for returns below 0.4 is due to bandlimited approximation. Because CVI reconstructs the distribution from a truncated frequency range $[-W, W]$ (see Table 1), it effectively acts as a low-pass filter. This is a fundamental trade-off of frequency-domain representations, smoothness comes at the cost of accurately representing sharp discontinuities.

(`is_slippery=True`). In this setting, the environment's stochasticity results in a highly sparse return distribution. Relying on the expected value can be misleading here, as it fails to capture the sharp distinction between reaching the goal and falling into a hole.

First, we verified the value convergence in this stochastic setting. Figure 3(b) shows that CVI converges to the optimal value alongside PI and VI.

We compared the learned return distributions against a Ground Truth distribution estimated using Monte Carlo rollouts. Initial attempts to visualize the probability density function (PDF) revealed challenges, as the ground truth distribution for this environment is highly sparse. To address this, we analyzed the cumulative distribution function (CDF), which offers a clearer view of the distributional match. Figure 4 presents the results for State 0. It clearly shows that CVI is able to approximate the ground-truth CDF smoothly, compared to C51 which produces a discrete representation. This is because CVI can operate in the continuous frequency domain, representing the distribution as a superposition of smooth sinusoidal waves. Unlike C51, which is structurally bound to a fixed, finite set of support atoms (forcing a step-function CDF). CVI's frequency representation allows it to model and reconstruct values continuously across the entire return space.

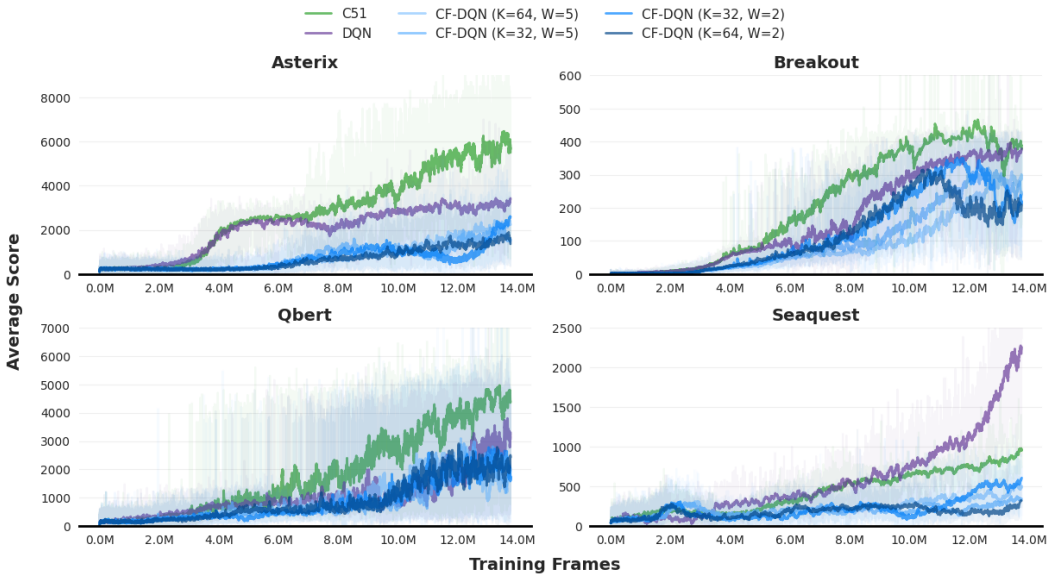


Figure 5: **Atari 2600 learning curves.** CF-DQN variants (four configurations varying $K \in \{32, 64\}$ and $W \in \{2, 5\}$) compared to DQN and C51 baselines over 40M frames. Curves show mean episodic returns across three seeds with EMA smoothing; shaded regions indicate variability. CF-DQN learns stably but underperforms baselines across all games, with no consistent hyperparameter winner.

5.2 Approximate Setting

We evaluate CF-DQN on four Atari 2600 games from the Arcade Learning Environment (Bellemare et al., 2013): Asterix, Breakout, Qbert, and Seaquest. These environments provide diverse challenges in terms of reward structure, temporal dependencies, and optimal policy complexity. We benchmark against two established baselines: DQN (Mnih et al., 2015) and C51 (Bellemare et al., 2017), both implemented within the CleanRL framework with identical hyperparameters and training procedures. All experiments are conducted over 3.5 million training steps (14 million environment frames with 4-frame skip) with three random seeds per configuration to assess statistical reliability.

To investigate the sensitivity of CF-DQN to frequency-domain hyperparameters, we evaluate four configurations varying two parameters: the number of frequency grid points $K \in \{32, 64\}$ and the maximum frequency range $W \in \{2, 5\}$. These settings correspond to different trade-offs between representational capacity and computational efficiency, with larger K enabling finer frequency resolution while larger W increases the bandwidth, allowing the model to capture high-frequency components that represent sharp discontinuities or narrow peaks in the return distribution.

Performance Analysis. Figure 5 presents the learning curves averaged across seeds, with shaded regions indicating variability. Our results reveal that while CF-DQN successfully learns non-trivial policies across all environments, demonstrating stable convergence and monotonic improvement, it consistently underperforms both baselines by substantial margins. Specifically, C51 achieves the highest scores on three of four games, while DQN dominates on Seaquest. In contrast, CF-DQN variants achieve 25-75% of baseline performance depending on the game and configuration.

Among CF-DQN configurations, no single hyperparameter setting demonstrates consistent superiority across environments. On Breakout, smaller frequency grids perform better, suggesting that excessive frequency resolution may introduce optimization challenges or overfitting in this sparse-reward setting. Conversely, on Qbert, all four CF-DQN variants converge to similar performance levels (2003-2094 range), with slightly higher means for configurations using $W{=}5$, indicating robustness to hyperparameter choice but limited benefit. On Seaquest, the $K{=}32, W{=}5$ configuration achieves the best CF-DQN result, though still substantially below baselines. Asterix exhibits the largest performance gap, with all CF-DQN variants scoring below 50% of DQN performance, suggesting particular sensitivity to the frequency-domain representation in this environment's reward structure.

5.3 Discussion

These results indicate that while the theoretical advantages of CVI, like exact distributional Bellman equations and simplified loss functions, translate successfully to tabular settings, the neural network function approximation regime presents additional challenges. complex-valued network outputs may suffer from optimization instabilities that are not present in real-valued architectures. Also, the polar interpolation and Gaussian collapse operations, while theoretically sound, introduce additional gradient paths that may disrupt credit assignment. On the other hand, the validity regularization term ($\lambda_{\phi_0} = 5.0$), while necessary to preserve the accuracy of the moment extraction, may constrain the representational flexibility of the network during training. Finally, the discretization of the frequency grid, highly precise in tabular settings, may interact poorly with neural network approximation errors.

The sensitivity to hyperparameters across environments further suggests that frequency-domain representations may require environment-specific tuning of K and W. Remarkably, the tabular experiments demonstrated that grid strategy choice was nearly irrelevant when using polar interpolation and Gaussian collapse. However, this robustness does not appear to transfer to the deep RL setting, where the interaction between discretization and function approximation introduces new failure modes.

6 Conclusion

Summary. In this work, we conducted a comprehensive implementation and evaluation of Characteristic Value Iteration, assessing its viability as a projection-free alternative for DistRL.

In the **tabular setting**, our results strongly validate the theoretical foundations laid by Farahmand (2019). We demonstrated that CVI is consistent with classical planning algorithms (VI and PI) in terms of expected value convergence. By operating in the continuous frequency domain, CVI demonstrated the ability to reconstruct smooth cumulative distribution functions (CDFs) without the quantization artifacts inherited by fixed-support methods like C51.

In the **approximate setting**, we successfully scaled the algorithm to high-dimensional state spaces through the proposed CF-DQN architecture. To the best of our knowledge, this represents the first implementation of CVI on the Atari benchmark. While CF-DQN demonstrated stable learning dynamics and the ability to acquire competent policies, it consistently underperformed established baselines like C51 and DQN. This performance gap highlights the non-trivial challenges of coupling frequency domain operations with deep neural networks.

Future Works. Our deep RL implementation relied entirely on design choices validated in tabular settings, being three-density-region grids, polar interpolation, and Gaussian collapse. However, the neural network approximation regime may require different configurations than the tabular case.

The approximate setting could benefit from an extensive ablation study which would help identify combinations better suited to gradient-based optimization with complex-valued networks, potentially closing or eliminating the performance gap with C51. Also, we used fixed hyperparameters ($K \in \{32, 64\}$, $W \in \{2, 5\}$) across all games. Simple heuristics based on reward statistics, such as setting W proportional to return variance, could improve performance without manual tuning per game.

We currently enforce $\tilde{V}(0; x, a) = 1$ through a soft penalty in the loss function, which competes with the Bellman objective during training. An alternative approach would use a normalization layer that guarantees this property by construction, eliminating the penalty term entirely and potentially stabilizing optimization. These directions suggest CF-DQN's current underperformance may be due to limited exploration of the design space rather than fundamental limitations of the frequency-domain approach.

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A CVI Hyperparameters

Table 1: Hyperparameters used for the Tabular CVI Experiments.

Hyperparameter	**Value**
General Settings	
Discount Factor (γ)	0.99
Convergence Threshold (ϵ)	1×10^{-6}
Max Iterations	200
Evaluation Episodes (Monte Carlo)	200
Max Steps per Episode	200
CVI-Specific Settings	
Grid Strategy	Two Density Regions
Frequency Range (W)	$[-20.0, 20.0]$
Number of Grid Points (K)	4096
Interpolation Method	Polar
Collapse Method	Gaussian